Ahmedabad University

CSE 631: Machine Learning

Weekly Report(1)

(Group - 9) Learning Machine

Section 1

Submitted to faculty: Prof. Mehul Raval

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Weekly Progress Report - 4

Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

1. Detailed Summary of Work Done During the Current Week

The fourth week of the project was mostly devoted to the annotation of the dataset and its preparation to supervised learning. The project proceeded in the same direction after the preprocessing and feature extraction pipeline was developed last week, which included the generation of labeled training data needed to train detection models. The most important task that was completed this week was to manually annotate the decorative stones in embroidery pictures through an annotation platform. This is necessary since machine learning models need to have properly labeled data that would teach the model how to identify and distinguish between stone areas and other embroidery patterns.

1.1 Dataset Preparation and Image Standardization

The dataset of embroidery images was arranged and pre-processed first so that the labeling and model training stages could be consistent.

The preparation procedures were:

1. Image Selection: The selection of embroidery fabric images with a decorative stone used in set images was made on the dataset.
2. Image Resizing: All the pictures were resized into 640 x 640 pixels because the pictures were to be the same input size across the dataset.
3. Dataset Organization: The images were organized in systematic folders, which could be easily uploaded and handled during the annotation.
4. Quality Checking: Up to adequate level of clarity and visibility of decorative stones, images were examined visually.

The image size is also standardized to ensure that the spatial representation is constant, and that advances the accuracy of the training object detection models.

1.2 Annotation Platform Setup

In order to conduct proper labeling of stone areas, an annotation tool was chosen and made to suit the project.

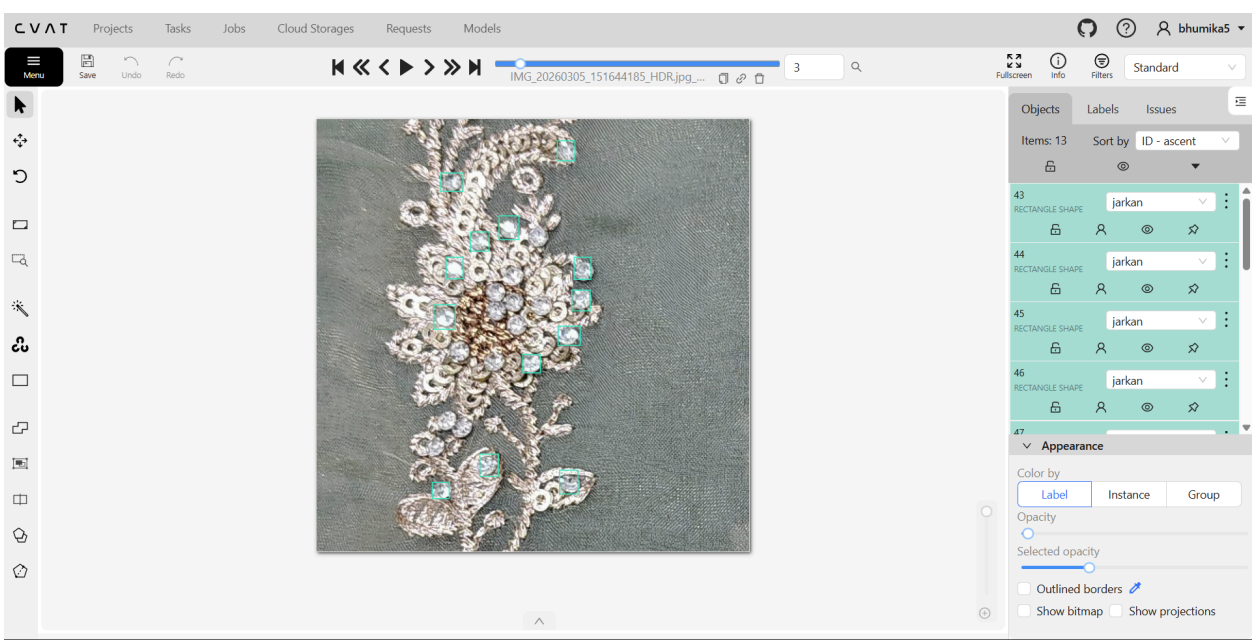
The labeling platform was CVAT (Computer Vision Annotation Tool) an open source annotation platform that is common among computer vision applications.

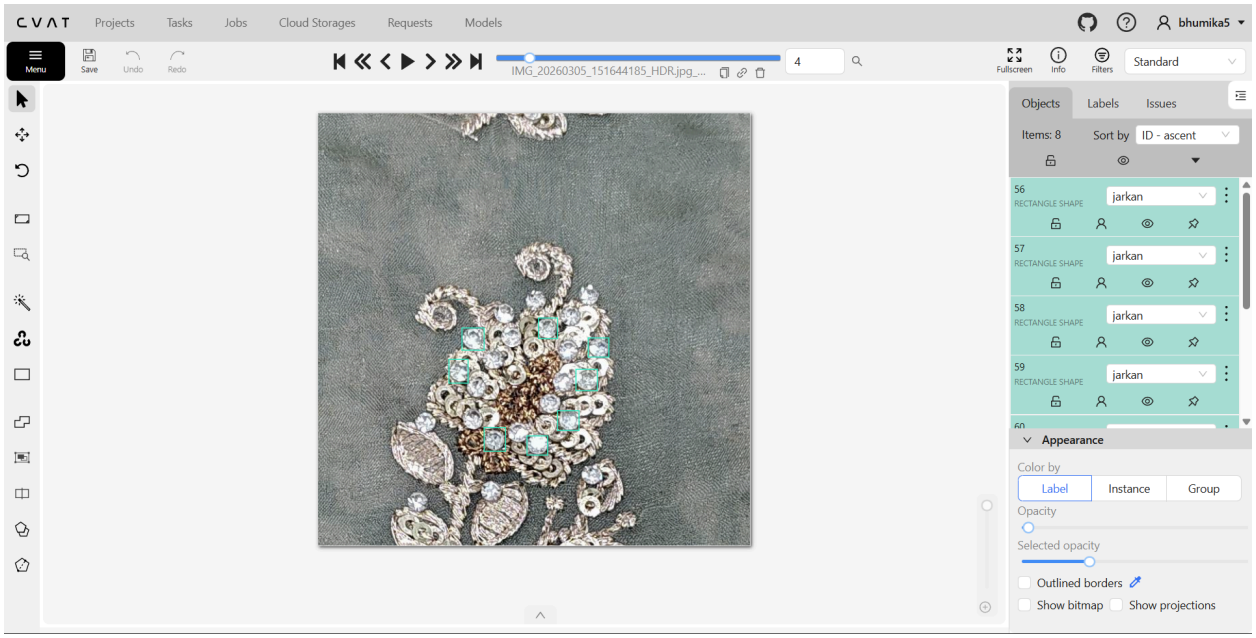
The setup process involved:

1. The establishment of a project space at CVAT.
2. Posting the ready embroidery pictures.
3. Specifying labels of annotations of the dataset.

The main tag developed on the project was: jarkan (Stone), this is the symbol of decorative stones found on embroidery images.

The features of CVAT that make it efficient in its annotation include bounding boxes, zooming, and navigating images and they prove handy in marking small objects such as stones.





1.3 Manual Annotation of Decorative Stones

Essentially, the most critical task of this week involved the manual coding of stone areas on the embroidery images.

All the pictures added to the annotation site were scrutinized closely and the decorative stones identified with bounding boxes. The labeling steps involved the following steps:

- Zooming in areas where there were stones.
- Visible stones: Drawing bounding boxes around each of them.
- Labeling each of the marked regions as Stone.
- Checking annotations to make sure that the enclosed stone area was well covered by the bounding box.

During labeling, special focus was needed as the embroidery images have complicated textures, threads, and reflections, and can make the identification of the stones difficult. It is important to ensure that annotation is correct since the accuracy of labels directly influences the performance of the trained model.

1.4 Annotation Verification and Dataset Quality Control

Following the first annotation stage, a verification process was performed to make sure that the quality and consistency of the labeled dataset are correct.

These steps included:

- Annotated images Reviewing them to identify missing or mis-labeled images.
- Making sure that the visible stones were well marked.
- Recalibrating the bounding boxes that failed to cover the stone regions.
- Standardization of labeling of all images.

This validation step aided in the assurance that the dataset was going to be effective and would be applicable in the training of machine learning frameworks.

1.5 Exporting the Annotated Dataset

After the labeling process had been done on the available images, the annotated dataset was exported out of the annotation platform to be used further in training the models.

The export process involved:

- Obtaining the annotation files of CVAT.
- Categorizing the labels into a system that can be used by machine learning models.
- Matching the annotation files to their respective images.

The coordinates of the bounding boxes of the locations of stones will be stored in these annotation files, and will serve as ground-truth data at the training phase.

1.6 Visualization and Annotation Inspection

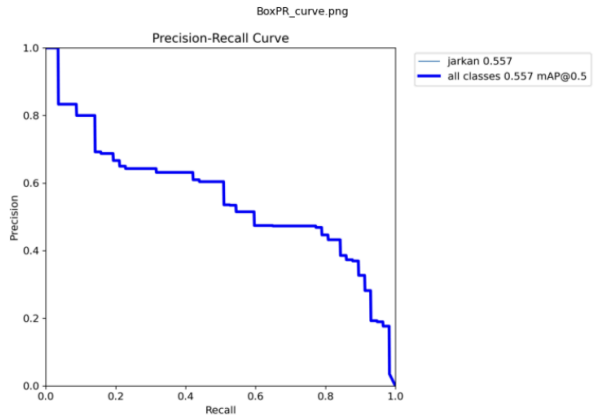
To confirm that the process of labeling was correct, a number of visualization checks were conducted.

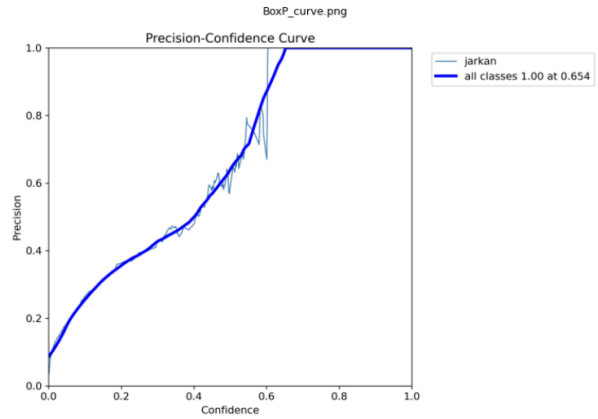
These included:

- Showing annotated images in bounding boxes.
- Checking the presence of stones in the bounded areas.
- Reviewing several sample images to identify the consistency in labeling.

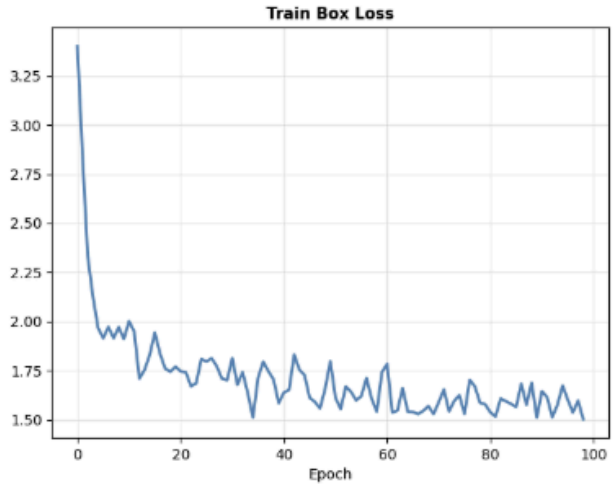
This verification served to make sure that the data was annotated correctly and is ready to be trained on stone detection models.

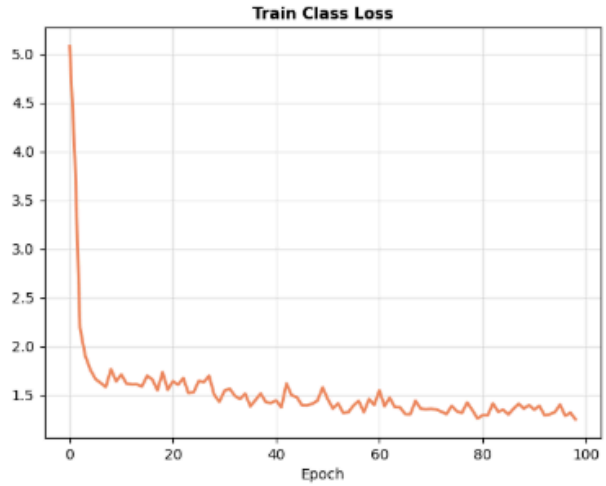
Sample image:

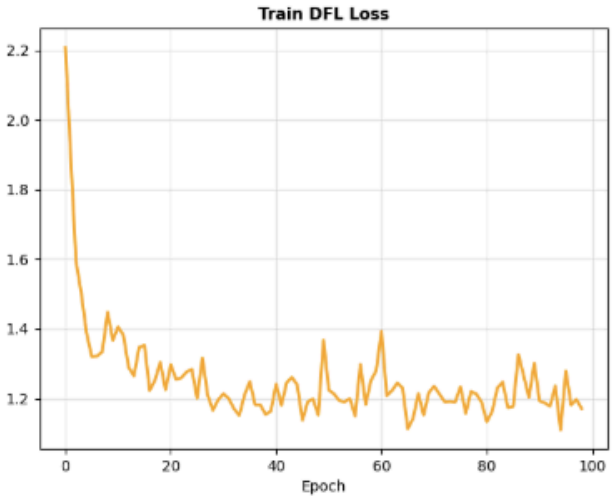


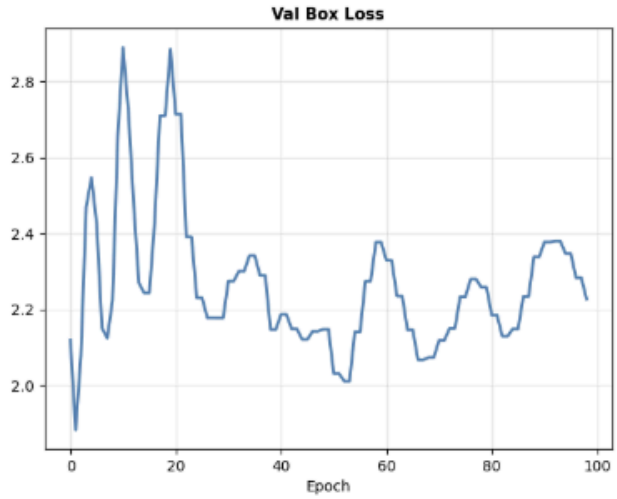


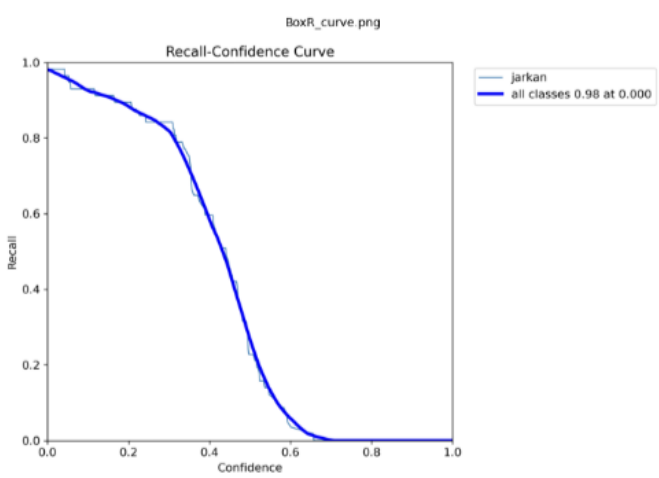
TOTAL JARKAN STONES DETECTED: 745

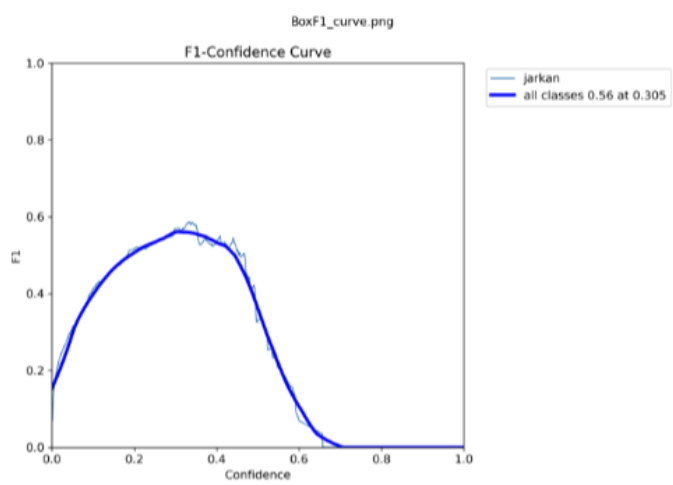

Train Box Loss
3.25
3.00
2.75
2.50
2.25
2.00
1.75
1.50
0
20
40
60
80
100
Epoch

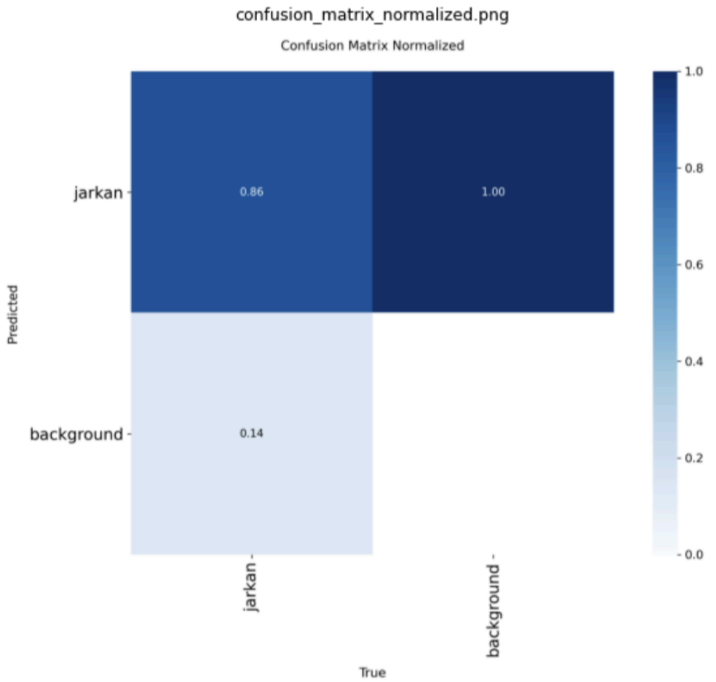

Train Class Loss
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
0
20
40
60
80
100
Epoch


Train DFL Loss
2.2
2.0
1.8
1.6
1.4
1.2
0
20
40
60
80
100
Epoch


Val Box Loss
2.8
2.6
2.4
2.2
2.0
0
20
40
60
80
100
Epoch


BoxR_curve.png
Recall-Confidence Curve
jarkan
all classes 0.98 at 0.000
Recall
Confidence
0.0
0.2
0.4
0.6
0.8
1.0


BoxF1_curve.png
F1-Confidence Curve
jarkan
all classes 0.56 at 0.305
F1
Confidence
0.0
0.2
0.4
0.6
0.8
1.0



2. Activities to be undertaken in the Week ahead.

The following step of the project will involve the training and evaluation of the object detection models by the use of the annotated dataset. The image labeling experiments developed in this week will be the basis of supervised learning models that will be used to identify decorative stones automatically.

2.1 Dataset Conversion and Preparation

The resultant annotated data that has been exported out of the annotation platform shall be translated into a format that can be utilized by object detection frameworks.

This step will include:

- Creating annotations and picture sets and training them.
- Division of dataset into training, validation and testing sets.
- Changing the form of annotation files in accordance with the format demanded by the chosen detection model.

Data set preparation is performed properly to make sure that the training process is smooth and efficient.

2.2 Training Object Detection Models

The next will be to train machine learning models that will be able to identify the decorative stones in image embroideries automatically.

The objectives include:

- Roeing the labelled data into the detection model.
- Trying to train the model to learn the patterns and visuals of stones.
- model parameters to enhance the performance of the detection.

The trained model must be capable of automatic localization and identification of stones in embroidery pictures.

2.3 Model Evaluation

The detection model should be trained and its performance should be assessed after training by using standard measures like:

- Accuracy
- Precision
- Recall
- F1-score

Also, visual analysis will be conducted based on the comparison of the predicted bounding box and ground-truth annotations.

2.4 Automatic Stone Detection and Counting

After the model proves to be reliable in terms of detection performance, the subsequent module is going to be concerned with counting stones in embroidery images automatically.

This module will:

- Find stone areas with the trained model.
- Find the number of stones found.
- Visualize results of display detection on the image.

This feature will be the objective of the automated stone counting system.

2.5 Result Visualization and Performance Analysis

In order to learn the behavior of the models and the performance of the system, various visualization tools will be used:

- Bounding Boxes of detected results images.
- Charts of performance comparison.
- Visualization of error analysis.

These illustrations will assist to examine the efficiency of the offered approach and define the potential improvements.